

resulting in a shorter instrument, but it is arguably more important for future investigations to focus on the reasons behind these high correlations.

Some of the limitations of this study have already been noted, such as the correlational design precluding any strong inferences about the nature of classroom instruction in fostering epistemological development. There may be something powerful in hands-on and inquiry-oriented instruction that promotes epistemological development, but this hypothesis awaits further study. The study also had a relatively small sample size and some of the cells in the three-way interactions (gender by ethnicity by SES) had few students in them, as discussed in the results section. Appropriate adjustments were made for the nonorthogonal design, but the power to detect reliable interactions was limited in some cells. We attempted to parse out effects due to ethnicity from those due to SES, by considering both variables in our analyses and by using a diverse sample, but race and ethnicity were still confounded to a degree. Larger and more diverse samples are needed to replicate our findings. Finally, the use of self-report instruments that focus students on specific aspects of epistemological beliefs may draw out more sophisticated beliefs than general interviews or discourse analysis of classroom language use (Hammer & Elby, 2002), although previous research has shown that there is at least a fair amount of agreement between interviews and questionnaires (Elder, 2002). There is a clear need for multi-trait, multi-method studies in this area to clarify the potential method variance artifacts from actual developmental differences in epistemological beliefs (Pintrich, 2002).

In summary, this study shows that young children have epistemological beliefs about science, at least in terms of their beliefs about the source, certainty, development, and justification for knowledge. These beliefs do show some change over time, even a relatively short period of time such as a nine-week science unit. More importantly, the results suggest that gender and ethnicity do not play as large a role in epistemological thinking and beliefs as achievement and SES. It seems likely that teachers and classrooms can influence the development of these beliefs and our results imply that one source of change might be the nature of hands-on and inquiry-oriented science instruction.

Appendix A. Self-report items used to measure epistemological beliefs

Source

Everybody has to believe what scientists say.

In science, you have to believe what the science books say about stuff.

Whatever the teacher says in science class is true.

If you read something in a science book, you can be sure it's true.

Only scientists know for sure what is true in science.

Certainty

All questions in science have one right answer.

The most important part of doing science is coming up with the right answer.

Scientists pretty much know everything about science; there is not much more to know.

Scientific knowledge is always true.

Once scientists have a result from an experiment, that is the only answer.

Scientists always agree about what is true in science.

Development

Some ideas in science today are different than what scientists used to think.

The ideas in science books sometimes change.

There are some questions that even scientists cannot answer.

Ideas in science sometimes change.

New discoveries can change what scientists think is true.

Sometimes scientists change their minds about what is true in science.

Justification

Ideas about science experiments come from being curious and thinking about how things work.

In science, there can be more than one way for scientists to test their ideas.

One important part of science is doing experiments to come up with new ideas about how things work.

It is good to try experiments more than once to make sure of your findings.

Good ideas in science can come from anybody, not just from scientists.

A good way to know if something is true is to do an experiment.

Good answers are based on evidence from many different experiments.

Ideas in science can come from your own questions and experiments.

It is good to have an idea before you start an experiment.

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